DATA VISUALIZATION

ggplot2 & Grammar of Graphics

CONTENT

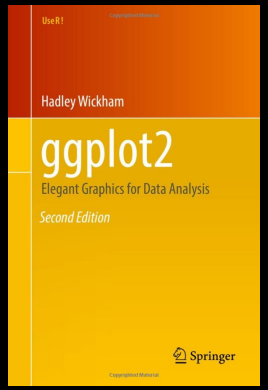
- Credits & References
- Pleas for Data Visualization
- The Grammar of Graphics
  - Basic Layers (data, aesthetics, geometry)
  - Advanced Layers (statistics, scales, facets, coordinates, themes)
- What to plot? Important visualizations for different applications

CREDITS

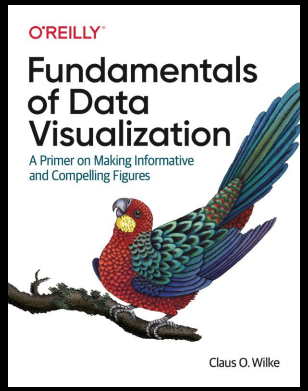
This slide deck is heavily inspired by the workshop “**Plotting anything with ggplot2**” by Tomas Lin Pedersen:

- Workshop video part 1
- Workshop video part 2
- Slides

REFERENCES



Wickham, Hadley. ggplot2. Springer Science + Business Media, LLC, 2016. Online verfügbar: https://ggplot2-book.org/



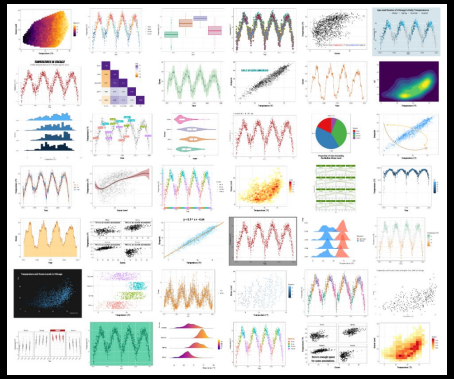
Wilke, C. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures. First edition, O'Reilly Media, 2019.

Online verfügbar: https://clauswilke.com/dataviz/index.html

REFERENCES

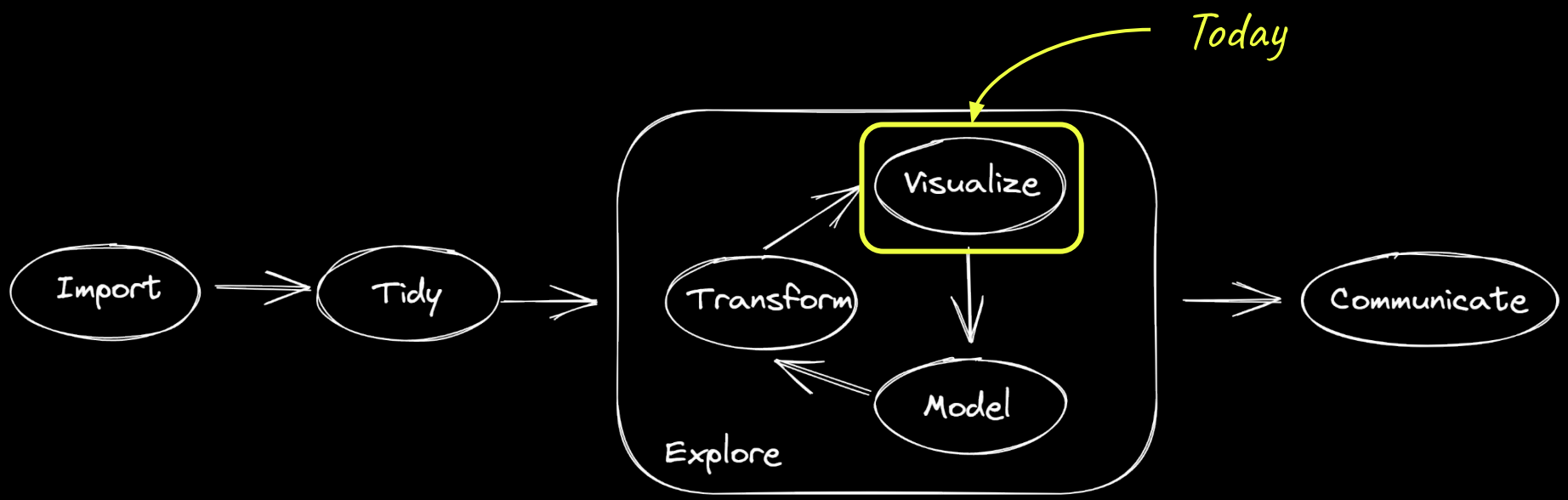
A `ggplot2` Tutorial for Beautiful Plotting in R

https://cedricscherer.netlify.app/2019/08/05/a-ggplot2-tutorial-for-beautiful-plotting-in-r/



DATA ANALYTICS PROCESS

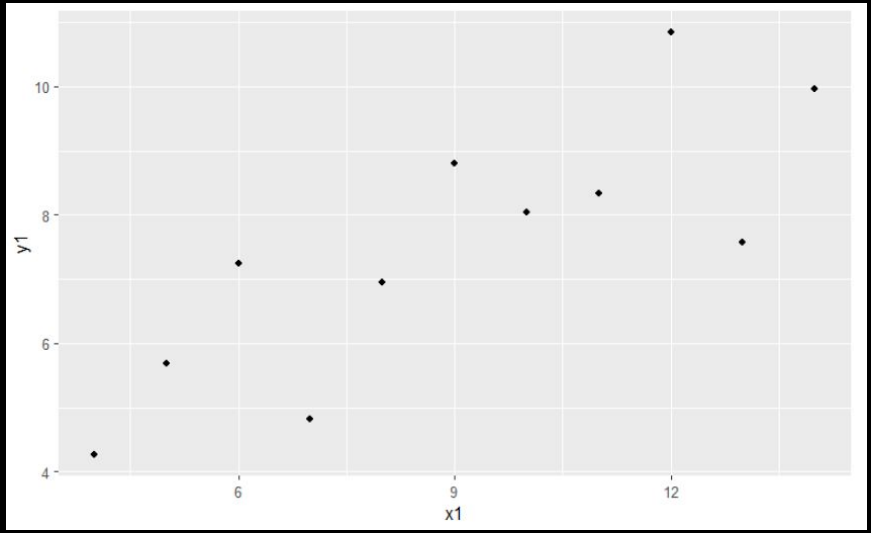
DATA VISUALIZATION



PLEAS FOR DATA VISUALIZATION

PLEAS FOR DATA VISUALIZATION

- Find two examples here



THE GRAMMAR OF GRAPHICS

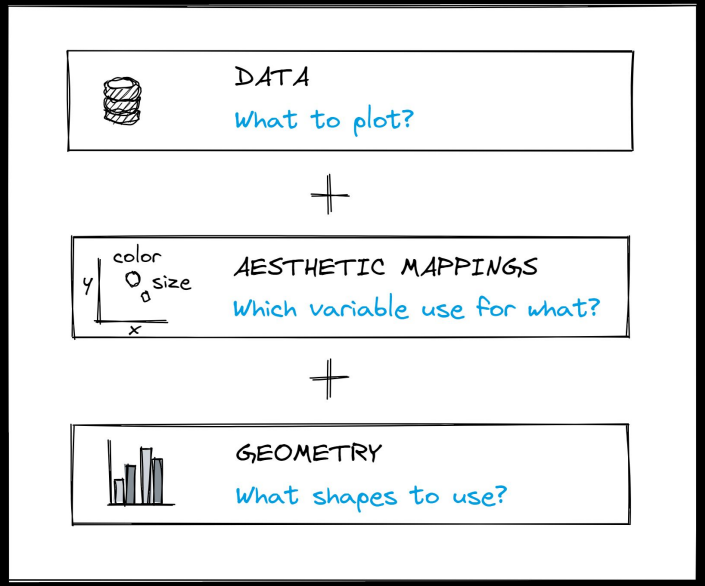
BASIC LAYERS

THE GRAMMAR OF GRAPHICS

BASIC LAYERS

Read more: https://ggplot2-book.org/layers.html

- In the Grammar of Graphics, a visualization consists of a minimum of three layers:
  - Data
  - Mapping of data to aesthetic elements
  - Geometric shapes
- `ggplot2` implements this idea → Visualizations are built as a stack of theses layers



THE GRAMMAR OF GRAPHICS

EXAMPLE FOR BASIC LAYERS

```
ggplot(covid) +
  aes(x = date, y = new_cases_smoothed_per_million) +
  geom_line()
```

What is the data?

Which geometric shape represents our data?

How to map the data to aesthetics?

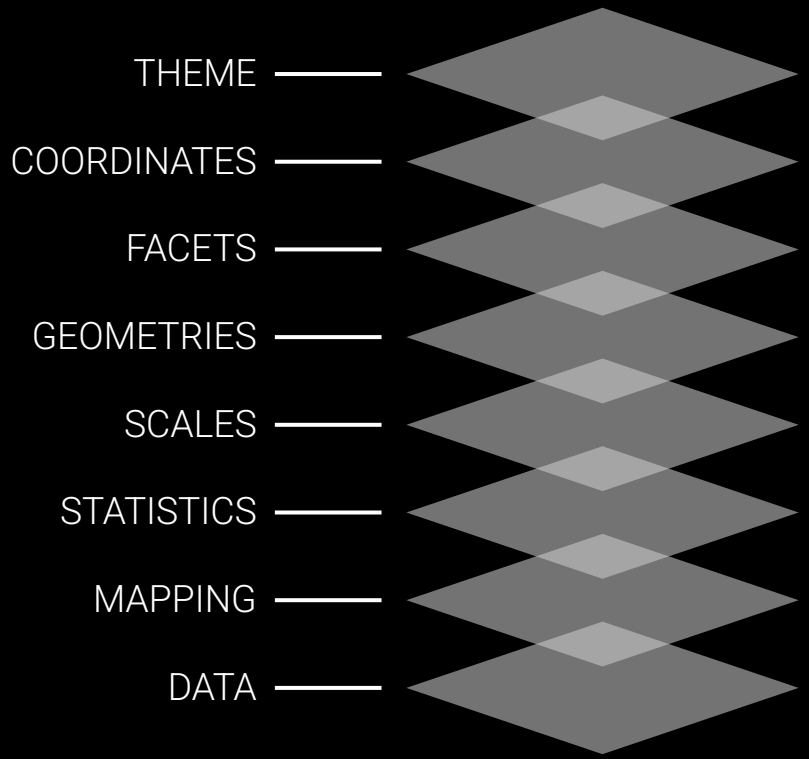
THE GRAMMAR OF GRAPHICS

ALL LAYERS

THE GRAMMAR OF GRAPHICS

COMPOSITION CONCEPT

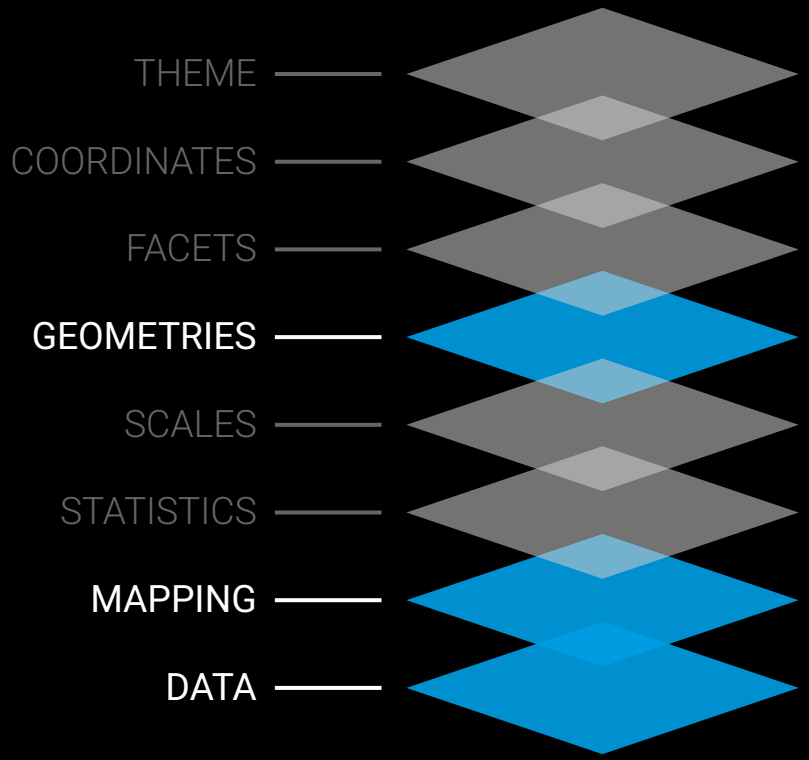
Any data visualization =



THE GRAMMAR OF GRAPHICS

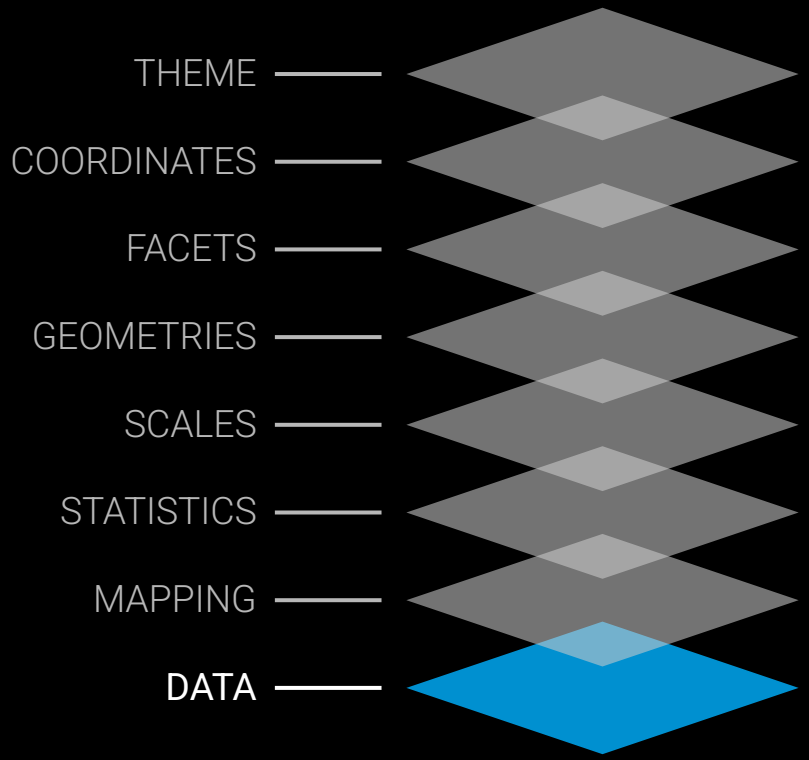
COMPOSITION CONCEPT

Only those 3 are needed! Everything else has a default!



THE DATA LAYER

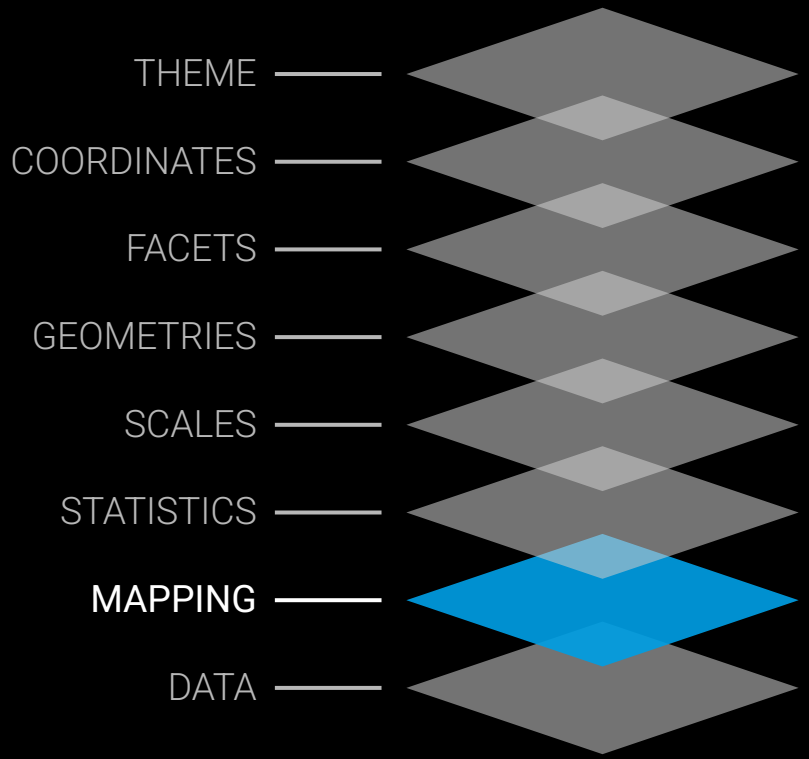
- Data must be provided as a data frame (`tibble`)
- Contains only
  - necessary variables
  - relevant rows and
  - the right level of aggregation
  - pre-computed statistics
- Toolset for data transformation (`dplyr`)



THE MAPPING LAYER

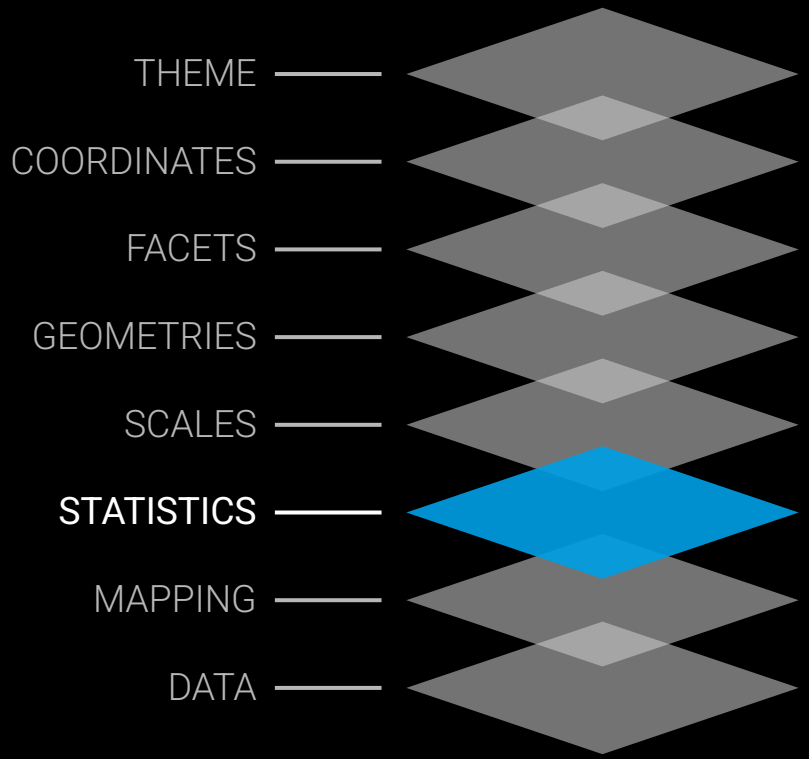
Check out: https://ggplot2tutor.com/aesthetics

- The aesthetics mapping (`aes`) links variables in the data to graphics properties
- Most important: What should be shown on x and y-axis?
- More mappings:
  - Line color & style
  - Fill color
  - Point size & shape
  - Alpha
  - …



THE STATISTICS LAYER

- If not pre-computed, statistics can be calculated by the visualization
- All geometries are assigned a default statistic (and vice versa)
- Example statistics:
  - `identity` → The value provided as is
  - `count` → Count rows
  - `bin` → Bin continuous variables
  - `density` → Estimate density
  - Many more…

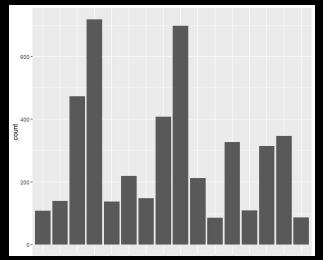


THE STATISTICS LAYER

STATS & GEOMS

```
tweets %>%
  ggplot() +
  stat_count(aes(x = screen_name))
```

```
tweets %>%
  ggplot() +
  geom_bar(aes(x = screen_name))
```



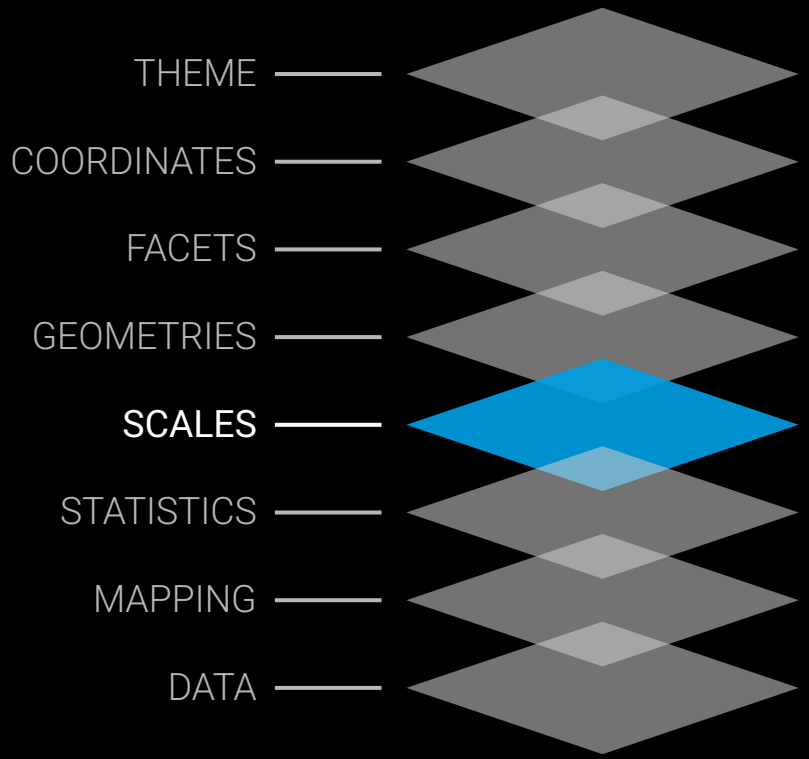
The count statistics uses bars per default

The bar geometry uses the count statistic per default

THE SCALES LAYER

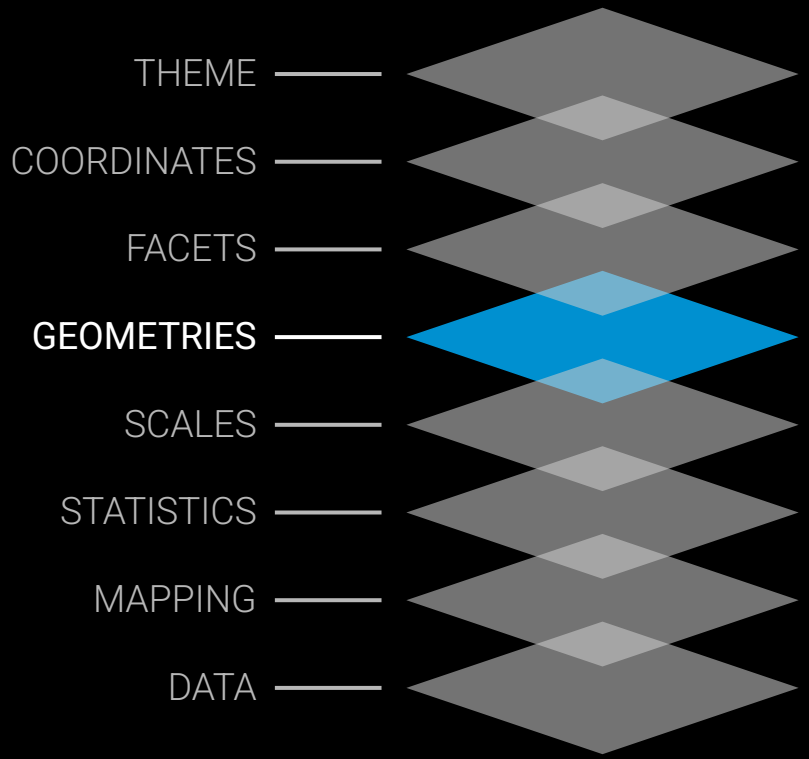
Check out: https://ggplot2tutor.com/scales

- All aesthetics mappings have a **scale** attached
- A **scale** maps values in the data to the x and y-axis, colors or sizes for shapes
- All scale functions follow the same naming scheme:
  - `scale_<aes>_<type>()`
- We use scales mainly for:
  - Color palettes
  - Axis labeling (breaks, formatting)



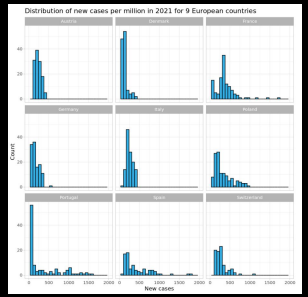
THE GEOMETRY LAYER

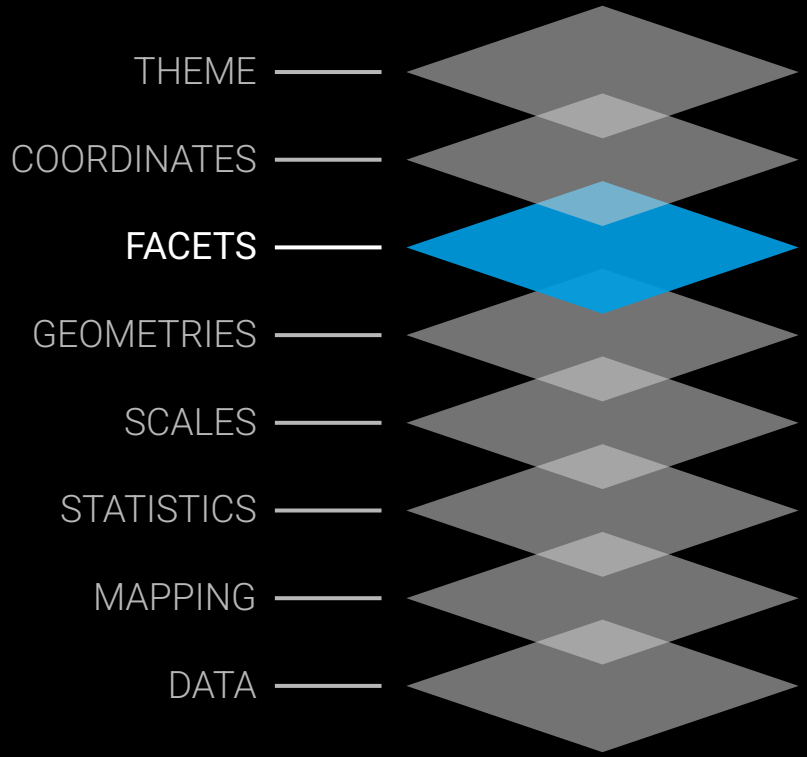
- The geometry is central to how the plot visualizes data
- Depending on the geometry, different aesthetics can or must be mapped
- We can add more than one geometry to a plot
- `geom_<type>()`



THE FACETS LAYER

- Create small panels with the same visualizations for different data
- Panel logic determined by variable in the data
- Good to avoid overplotting and maintain readability!
- `facet_wrap()` vs `facet_grid()`

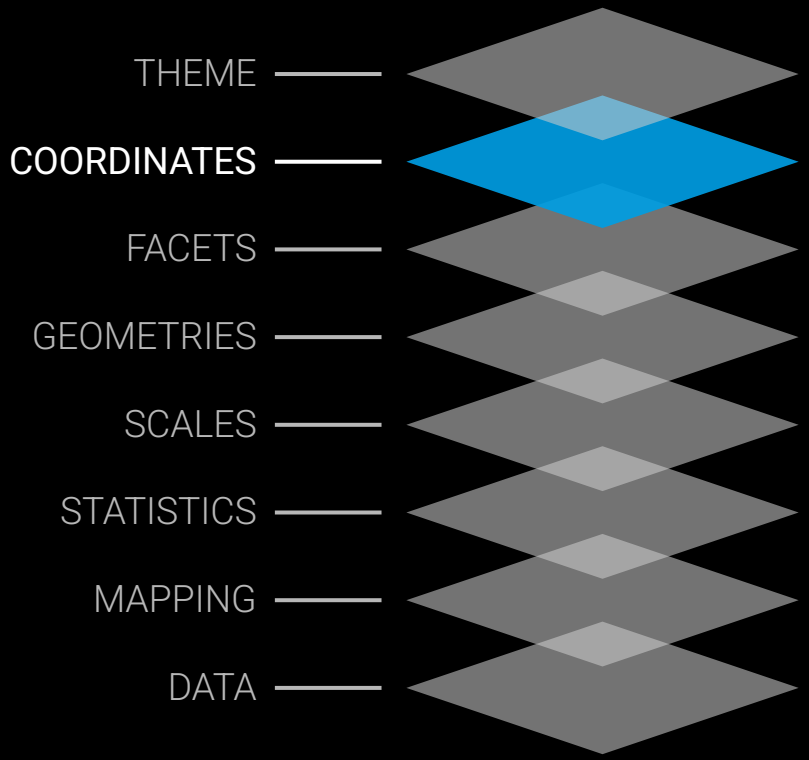




THE COORDINATES LAYER

- Specify the coordinate system underlying the visualization:
  - Cartesian (default)
  - Polar
- Allows for changing axis limits (just like scales)
- `coord_flip()` is useful to quickly flip x and y

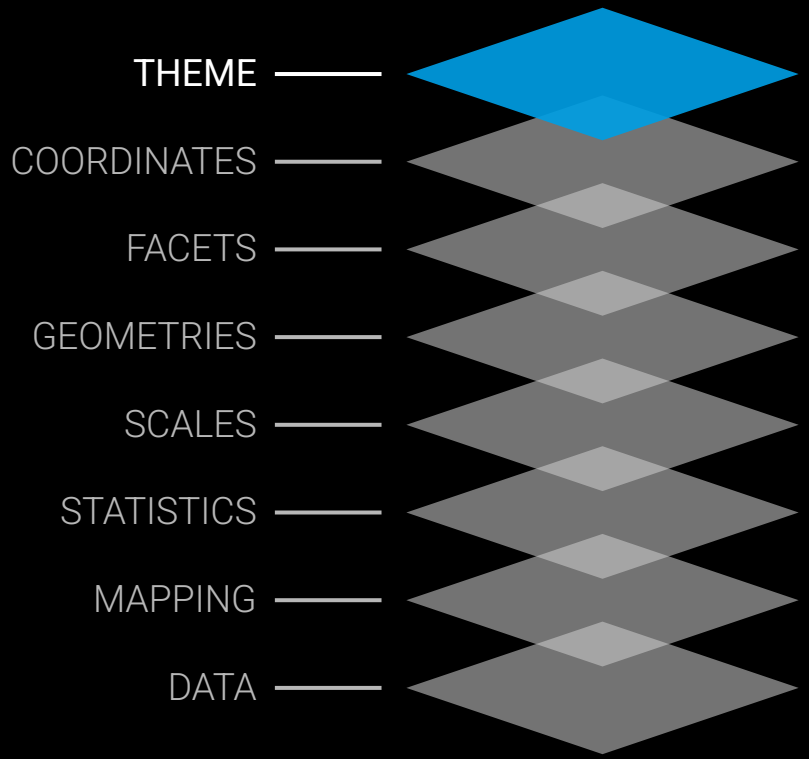
We will rarely use this layer!



THE THEME LAYER

Check out: https://ggplot2tutor.com/theme

- Style the plot
  - Background colors
  - Fonts (axis, titles)
  - Legends
  - …
- There are predefined themes for us to use:
  - `theme_bw()`
  - `theme_light()`
  - `theme_dark()`



WHAT TO PLOT?

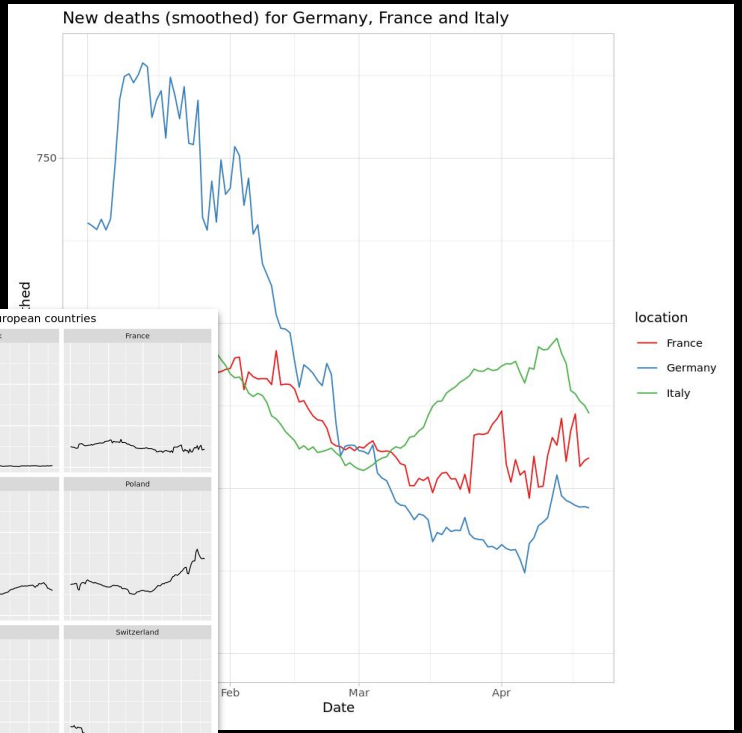
WHAT TO PLOT?

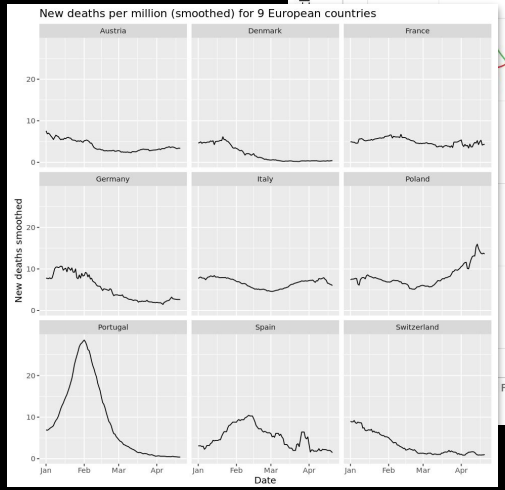
TRENDS & DEVELOPMENTS

Read more: https://clauswilke.com/dataviz/time-series.html

- x-axis displays the time (usually), the y-axis some value over time:
  - Line chart
  - Area Chart
  - One vs. multiple series
  - Facets

- Example: Covid19



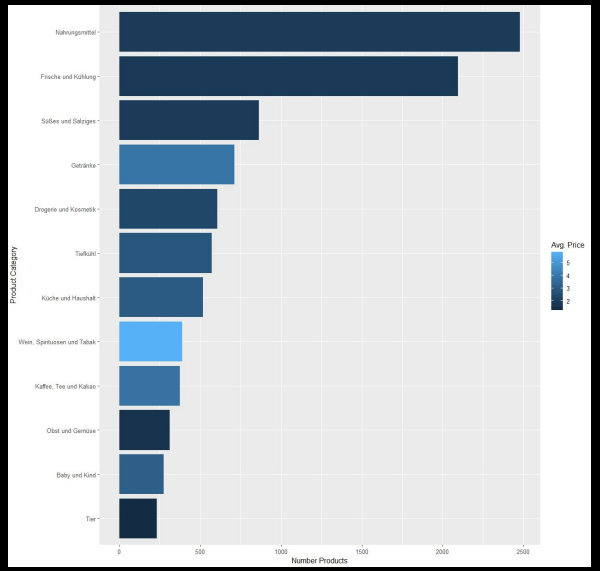


WHAT TO PLOT?

AMOUNTS & PROPORTIONS

Read more:
https://clauswilke.com/dataviz/visualizing-proportions.html

- A geometry's size (height, width, area) represents values in the data for easy comparison:
  - Bar Chart
    - next to each other
    - stacked
  - ~~Pie chart~~

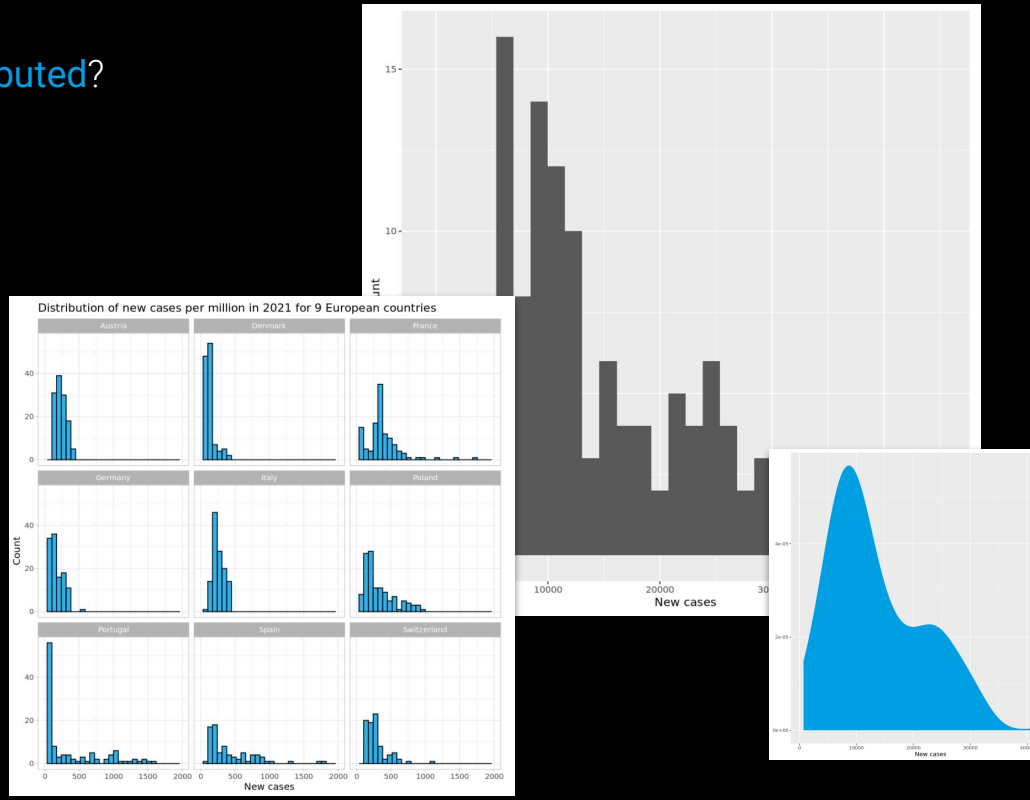


WHAT TO PLOT?

DISTRIBUTIONS

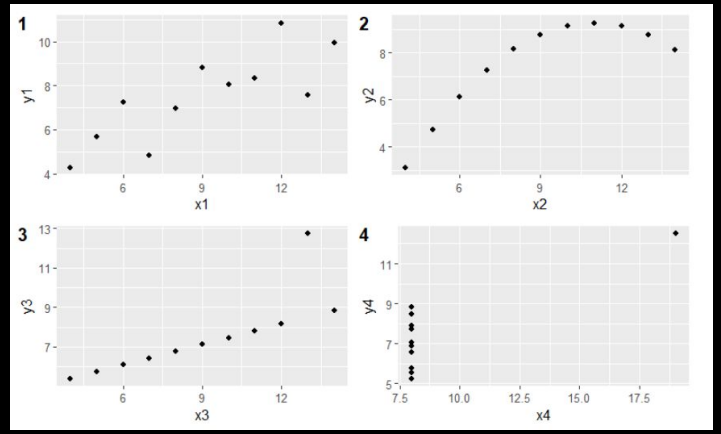
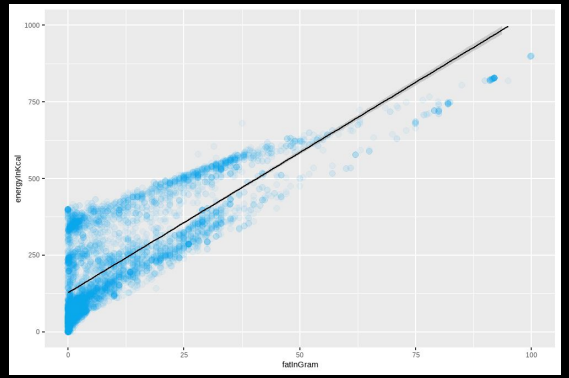
Read more:
https://clauswilke.com/dataviz/histograms-density-plots.html

- How are observations of a variable distributed?
  - Histogram (one vs. multiple series)
  - Density plot
  - Ridgeline Plots
  - Box plots



WHAT TO PLOT?

ASSOCIATIONS

Read more:
https://clauswilke.com/dataviz/visualizing-associations.html

- What associations between variables can we find in the data?
  - Point diagram (*scatter plot*)
  - Trendlines
  - Heat maps